

OpenStax Calculus Volume 2

Section 4.5 Exercises: First-Order Linear Equations

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Are the following differential equations linear? Explain your reasoning.

1. $\frac{dy}{dx} = x^2y + \sin x$

2. $\frac{dy}{dt} = ty$

3. $\frac{dy}{dt} + y^2 = t$

4. $y' = x^3 + e^x$

5. $y' = y + e^y$

Write the following first-order differential equations in standard form.

6. $y' = x^3y + \sin x$

7. $y' + 3y - \ln x = 0$

8. $-xy' = (3x + 2)y + xe^x$

9. $\frac{dy}{dt} = 4y + ty + \tan t$

10. $\frac{dy}{dt} = yt(t + 1)$

What are the integrating factors for the following differential equations?

11. $y' = xy + 3$

12. $y' + e^xy = \sin x$

13. $y' = x \ln(x)y + 3x$

14. $\frac{dy}{dx} = \tanh(x)y + 1$

15. $\frac{dy}{dt} + 3ty = e^ty$

Solve the following differential equations by using integrating factors.

16. $y' = 3y + 2$

17. $y' = 2y - x^2$

18. $xy' = 3y - 6x^2$
19. $(x + 2)y' = 3x + y$
20. $y' = 3x + xy$
21. $xy' = x + y$
22. $\sin(x)y' = y + 2x$
23. $y' = y + e^x$
24. $xy' = 3y + x^2$
25. $y' + \ln x = \frac{y}{x}$

Solve the following differential equations. Use your calculator to draw a family of solutions. Are there certain initial conditions that change the behavior of the solution?

26. [T] $(x + 2)y' = 2y - 1$
27. [T] $y' = 3e^{t/3} - 2y$
28. [T] $ty' + \frac{y}{2} = \sin(3t)$
29. [T] $xy' = 2\frac{\cos x}{x} - 3y$
30. [T] $(x + 1)y' = 3y + x^2 + 2x + 1$
31. [T] $\sin(x)y' + \cos(x)y = 2x$
32. [T] $\sqrt{x^2 + 1}y' = y + 2$
33. [T] $x^3y' + 2x^2y = x + 1$

Solve the following initial-value problems by using integrating factors.

34. $y' + y = x, y(0) = 3$
35. $y' = y + 2x^2, y(0) = 0$
36. $xy' = y - 3x^3, y(1) = 0$
37. $x^2y' = xy - \ln x, y(1) = 1$
38. $(1 + x^2)y' = y - 1, y(0) = 0$
39. $xy' = y + 2x \ln x, y(1) = 5$
40. $(2 + x)y' = y + 2 + x, y(0) = 0$
41. $y' = xy + 2xe^{\frac{1}{2}x^2}, y(0) = 2$
42. $\sqrt{xy}' = y + 2x, y(0) = 1$
43. $y' = 2y + xe^x, y(0) = -1$
44. A falling object of mass m can reach terminal velocity when the drag force is proportional to its velocity, with proportionality constant k . Set up the differential equation and solve for the velocity given an initial velocity of 0.

45. Using your expression from the preceding problem, what is the terminal velocity? (Hint: Examine the limiting behavior; does the velocity approach a value?)
46. [T] Using your equation for terminal velocity, solve for the distance fallen. How long does it take to fall 5000 meters if the mass is 100 kilograms, the acceleration due to gravity is 9.8 m/s^2 and the proportionality constant is 4?
47. A more accurate way to describe terminal velocity is that the drag force is proportional to the square of velocity, with a proportionality constant k . Set up the differential equation and solve for the velocity.
48. Using your expression from the preceding problem, what is the terminal velocity? (Hint: Examine the limiting behavior: Does the velocity approach a value?)
49. [T] Using your equation for terminal velocity, solve for the distance fallen. How long does it take to fall 5000 meters if the mass is 100 kilograms, the acceleration due to gravity is 9.8 m/s^2 and the proportionality constant is 4? Does it take more or less time than your initial estimate?

For the following problems, determine how parameter a affects the solution.

50. Solve the generic equation $y' = ax + y$. How does varying a change the behavior?
51. Solve the generic equation $y' = ay + x$. How does varying a change the behavior?
52. Solve the generic equation $y' = ax + xy$. How does varying a change the behavior?
53. Solve the generic equation $y' = x + axy$. How does varying a change the behavior?
54. Solve $y' - y = e^{kt}$ with the initial condition $y(0) = 0$. As k approaches 1, what happens to your formula?